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RHEUMATIC DISORDERS OF CHILDHOOD

Juvenile Dermatomyositis (JDM)

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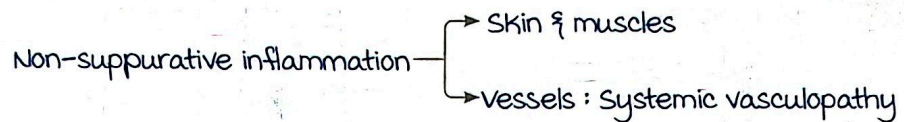
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multisystem disorder.

m/c cause of inflammatory myopathy in children.

Age group : 5-14 yrs (F > m).

PATHOGENESIS :



CLINICAL FEATURES :

1. Proximal, symmetrical muscle weakness :

- Involves shoulder & hip muscles (Early feature) → Difficulty in getting up from a chair/climbing stairs.
- **Gower's sign positive** : Using distal muscles (Eg : Hand) as support over proximal muscles (Eg : Thigh) to get up.

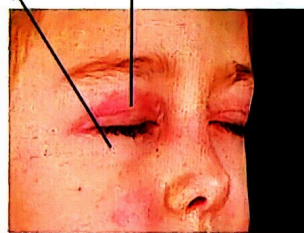
a. Skin findings :

Red, scaly lesions over extensor aspect of metacarpophalangeal & interphalangeal joints

Violaceous discoloration
Periorbital edema



Gottron papules



Heliotrope rash



Gottron papules



Shawl sign

3. Nail Findings :

Periungual erythema.

Dilated nail fold capillaries on nail fold capillaroscopy.

DIAGNOSIS :**Essential Criteria :**

1. **Rash** : Heliotrope rash & gottron's papules.

Other Criteria :

Any 3 of the following :

1. muscle weakness : Proximal & symmetrical.
 2. Elevated muscle enzymes : CPK, Aldolase, LDH.
 3. Electromyogram (EMG) : Fasciculations, high-frequency discharges.
 4. muscle biopsy : Lymphocytic infiltrates.
- } Replaced by MRI in clinical practice

Note :

myositis specific antibodies : used to monitor **prognosis**.

- Anti-TNF γ antibodies : Severe skin involvement; refractory to Rx.
- Anti-mi-2 antibodies : ~~muscle involvement~~ good response to Rx.

TREATMENT :**Steroids :**

- Initial treatment.
- IV methylprednisolone.

methotrexate (DMARD) :

- maintenance therapy : Weekly for 2 years.

Taper with oral prednisolone

Juvenile Idiopathic Arthritis (JIA)

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most common chronic rheumatic arthritis in children.

Autoimmune disorder.

DEFINITION :

- Age of onset < 16 years.
- Duration $\geq$ 6 weeks.
- Involvement of at least 1 joint.
- Exclusion of other conditions.

TYPES :**1. Oligoarticular JIA :**

- m/c type (F > m).

$\leq$ 4 joints involvement.

Large joints affected : **knee** > Ankle > Wrist.

Extra-articular findings : **uveitis**

- Strongly a/w **positive ANA**.

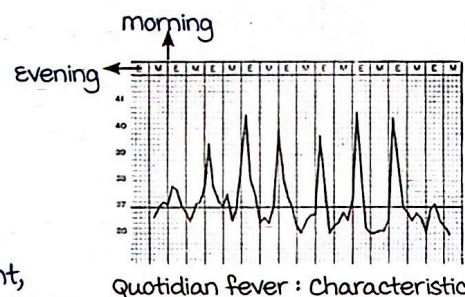
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2. Polyarticular JIA :

	Rheumatoid factor -	Rheumatoid factor +
Age	Any age upto 16 years	9-11 years
Joints affected >4	- Knee. - Hip. - Wrist.	- metacarpophalangeal joint. - Proximal interphalangeal joint. - Cervical spine. - Temporomandibular joint.
Symmetry	Asymmetrical	Symmetrical
Association	Uveitis (ANA +)	Rheumatoid nodules (Elbow, occiput, achilles tendon)

3. Systemic JIA :

- m > f.
- Quotidian fever : 1-2 spikes/day.
- Arthritis.
- Any one of the following :
 - Salmon-coloured, non-pruritic, evanescent, maculopapular rash.
 - Hepatosplenomegaly.
 - Lymphadenopathy.
 - Serositis (Pleuritis/pericarditis).
- Koebner Phenomenon : Lesions along the line of superficial trauma.
 - Hypersensitivity reaction.



Salmon rash

Complication : Macrophage Activation Syndrome (MAS).

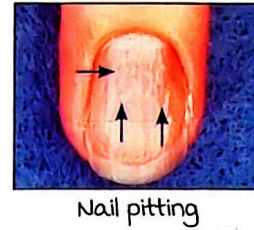
- Hyperinflammation (↑↑ production of interleukins & cytokines).
- Continuous fever.
- Drop in cell counts (Especially thrombocytopenia).
- Hypofibrinogenemia (D/t consumptive coagulopathy) → Low ESR.
- ↑ ferritin & ↑ triglycerides.
- Hemophagocytosis on bone marrow biopsy.

4. Psoriatic arthritis :

- F > m.
- Arthritis →
 - Psoriasis OR
 - Any 2 of the following :
 - Nail pitting.
 - Dactylitis (Inflammation of fingers : 'Sausage digits').
 - 1° relative with psoriasis.

5. Enthesitis related arthritis :

- m > F.
- Inflammation of tendons/ligaments.
- Asymmetrical arthritis of lower limb joints.
- **HLA-B27 association** → Can also present with :
 - Sacro-iliitis.
 - Precursor to ankylosing spondylitis.



Nail pitting

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INVESTIGATIONS :

JIA is a clinical diagnosis (Lab tests not required).

Inflammatory Evidence : Prognostic use.

- ↑ ESR
- ↑ CRP.

TREATMENT :

Type	Treatment
Oligoarticular	NSAIDs (Analgesia) If unresponsive Intra articular triamcinolone
Polyarticular	NSAIDs + DMARDs (methotrexate > Hydroxychloroquine)
Systemic	Biological agents : • Anakinra : IL-1 receptor antagonist. • Adalimumab, etanercept : Anti-TNF α antibody. • Tocilizumab : Anti-IL-6 antibody.

Henoch Schonlein Purpura (HSP)/IgA Vasculitis

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multisystem small vessel vasculitis.

m/c childhood vasculitis worldwide.

Age : 4-8 yrs.

PATHOGENESIS :

- **Leukocytoclastic vasculitis** (Deposition of neutrophil debris in inflamed blood vessel walls).
- Fibrinoid necrosis.
- IgA deposits.

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CLINICAL FEATURES :

Palpable purpura :

- m/c & characteristic feature.
- Non-blanching.
- Non-thrombocytopenic.
- Location :
 - Dependent : Lower limbs.
 - Genital area (males).



HSP

Note : Thrombocytopenic purpura can be seen all over the body.

Other Features :

- Abdominal pain : Post-prandial, intermittent, peri-umbilical, colicky.
 - Joint involvement : Arthralgia/arthritis in ankle & knee joint.
 - Renal involvement :
 - Hematuria/nephritis (30 % cases)
 - Nephrotic range proteinuria (10 % cases)
- } Long term renal complications.

INVESTIGATIONS :

Skin/kidney biopsy :

- usually not required (Clinical diagnosis).
- Leukocytoclastic vasculitis seen.
- IgA deposition on immunofluorescence.

DIAGNOSTIC CRITERIA :

Essential	Additional (At least one)	
• Raised, palpable, hemorrhagic skin lesions in the absence of thrombocytopenia.	• Abdominal pain. • Positive biopsy findings.	• Arthritis/arthralgia. • Renal involvement.

Note : Platelet count is normal/↑.

COMPLICATIONS :

Acute :

GI involvement : Intussusception, GI bleed.

Chronic :

Renal involvement :

- Age >6 yrs predisposed to long-term renal problems.
- Regular monitoring of renal function for at least 1 year.
- 1-5 % → Progressive kidney disease (CKD).

PROGNOSIS :

- Good prognosis.
- Long term follow-up if renal involvement.

TREATMENT :

- Supportive treatment (Self-limiting).
- **Steroids** : Severe GI involvement.
- 10-20 % have recurrence.

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Kawasaki Disease

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medium vessel vasculitis (Primarily coronary arteries).

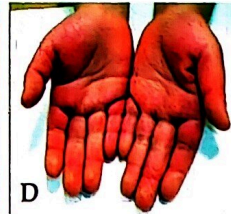
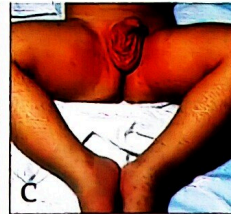
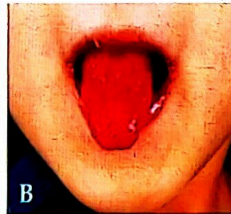
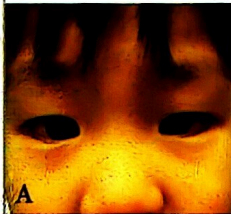
most common childhood vasculitis in India.

Age : < 5 years.

Idiopathic etiology.

STAGES AND CLINICAL MANIFESTATIONS :

Stage	Features
Acute (1-2 weeks) (↓ morbidity if diagnosed in this stage)	• > 5 days of fever (Without any discernible cause). - mnemonic : CREAM. • Conjunctivitis : B/I non-purulent, spares limbus. • Rash : Non-specific maculopapular rash. • Extremities : Non-pitting edema & erythema of limbs. • Lymphadenopathy : w/ cervical. • mucocutaneous involvement. - Oral cavity erythema → Lips : Cheilitis → Strawberry tongue - Cracking of lips.
Subacute : Complications (3-4 weeks) (most dangerous)	• Coronary aneurysm : - 25% of cases. - Predisposes to myocardial ischemia/thrombosis/rupture. • Periungual desquamation : - Skin peeling around nails. • Thrombocytosis.



Stage	Features
Convalescent (6-8 weeks)	• Stage of recovery. • Beau's lines on nails.

Feedback

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Other manifestations :

- BCG scar reactivation.
- Perianal desquamation.
- Aseptic meningitis : High irritability.
- Hydrops of gallbladder : Right hypochondrial pain.
- Urethritis : Sterile pyuria.
- Joint involvement : Arthralgia/Arthritis.



BCG scar reactivation



Perianal desquamation

DIAGNOSTIC CRITERIA (CLINICAL) :

- Fever ≥ 5 days
- +
 - 4 out of 5 CREAM findings.

INVESTIGATIONS :

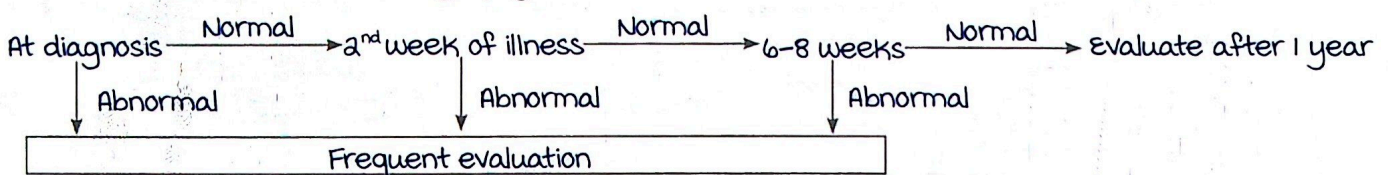
1. Blood count : Platelet count.

Normal : Acute phase Increased : Subacute phase

2. Inflammatory markers :

- $\uparrow$ CRP.
- $\uparrow$ ESR.

3. Echocardiography :



ATYPICAL/INCOMPLETE KAWASAKI DISEASE :

[Fever ≥ 5 days + 2/3 CREAM features] OR [Infant with fever ≥ 7 days & no other manifestations]

ESR ≥ 40 mm/hr.

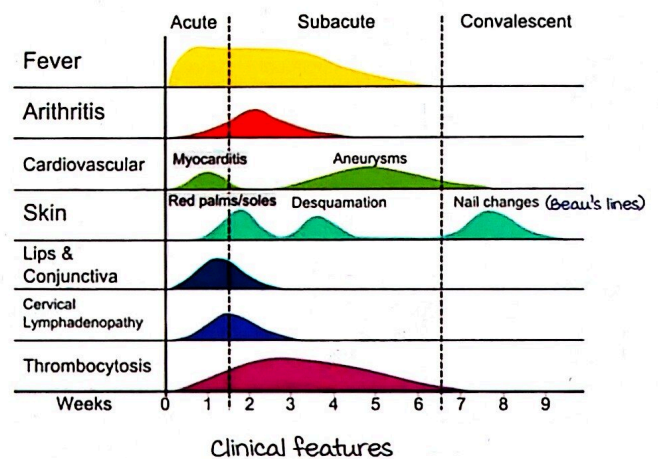
CRP ≥ 3 mg/dL.

Echocardiography

Cardiac involvement seen

Treat as Kawasaki disease

Clinical manifestations of Kawasaki Disease



TREATMENT :

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- Intravenous immunoglobulin (IVIg) :

- Treatment of choice.
- Dose : 2 g/kg.
- ↓ risk of coronary abnormalities to 2-3 %.

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- Aspirin :

